

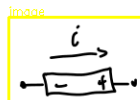
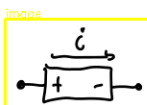
Rappel

$$q = 1.6022 \times 10^{-19} \text{ C}$$

$$q = 1.6022 \times 10^{-19} \text{ C}$$

$$V = \frac{dw}{dq}$$

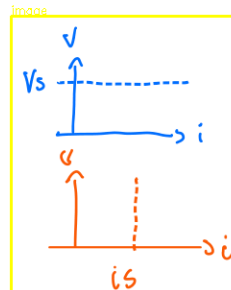
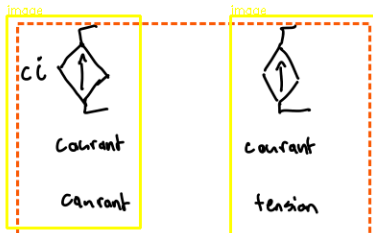
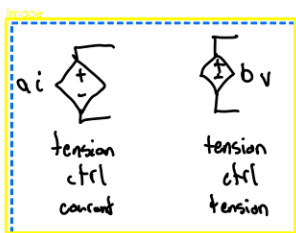
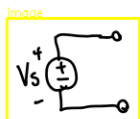
$$i = \frac{dq}{dt}$$



$$p = +i V$$

an

$$-i V$$

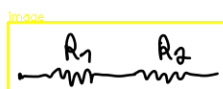
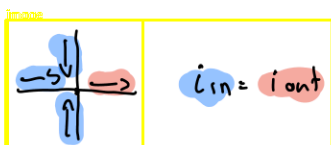


$$i = \frac{V}{R}$$

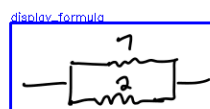
$$V = Ri$$

$$p = Ri^2 = \frac{V^2}{R}$$

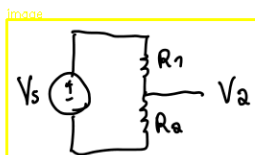
$$G = 1/R$$



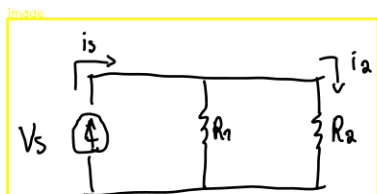
$$R = R_1 + R_2$$



$$R = \left(\frac{1}{R_1} + \frac{1}{R_2} \right)^{-1}$$



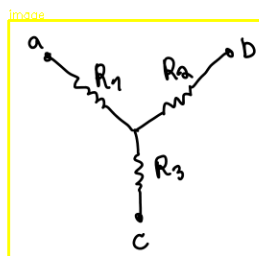
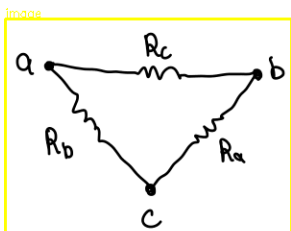
$$V_a = V_s \cdot \frac{R_a}{R_1 + R_a}$$



$$i_2 = i_s \cdot \frac{R_1}{R_1 + R_a}$$

Trans phorm

$$\Delta \rightarrow Y$$



$$R_{ab} = \frac{R_c(R_a + R_b)}{R_a + R_b + R_c} = R_1 + R_2$$

$$R_{ac} = \frac{R_b(R_a + R_c)}{R_a + R_b + R_c} = R_1 + R_3$$

$$R_{bc} = \frac{R_a(R_b + R_c)}{R_a + R_b + R_c} = R_2 + R_3$$

$$\Delta \rightarrow Y$$

$$Y \rightarrow \Delta$$

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_2}$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_3}$$

Vocabulaire :

text

Noeud : point où 2+ composants se rejoignent

Noeud essentiel : 3+ compo

text

parcours : chemin avec n° compo en double

text

Branche : parcours reliant 2 noeuds

text

Branche ess : parcours relie 2 noeuds essentiels sans passer par autre ess

Boucle : parcours fermé

text

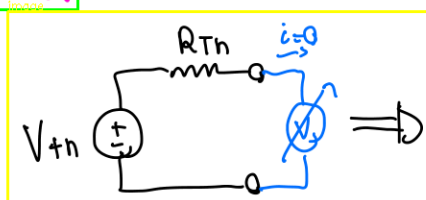
Maille : boucle sans autres boucle

text

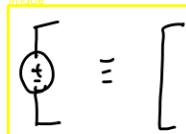
Thev

text

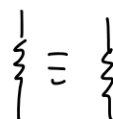
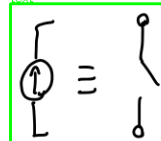
pour trouver r_{th} , mod circuit avec :



image



text

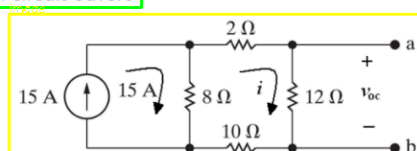


paragraph-title

Calculer l'équivalent de Thévenin - Exemple (1)

text

Tension en circuit ouvert



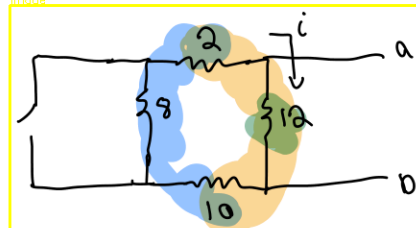
text

$$2i + 12i + 10i + 8(i - 15) = 0$$

$$\therefore i(2 + 12 + 10 + 8) = (8)(15) \Rightarrow i = 3.75 \text{ A}$$

$$v_{oc} = 12i = 45 \text{ V} = V_{Th}$$

image



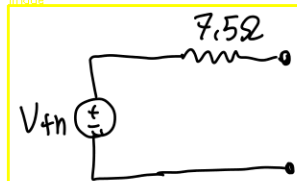
text

$$R = 20\Omega$$

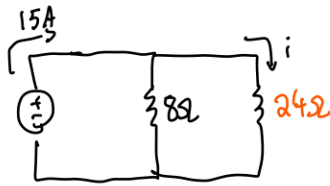
display formula

$$\left(\frac{1}{12} + \frac{1}{20}\right)^{-1} = 7.5 = R_{th}$$

image

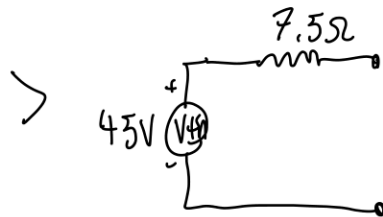


Transfer V_{th}



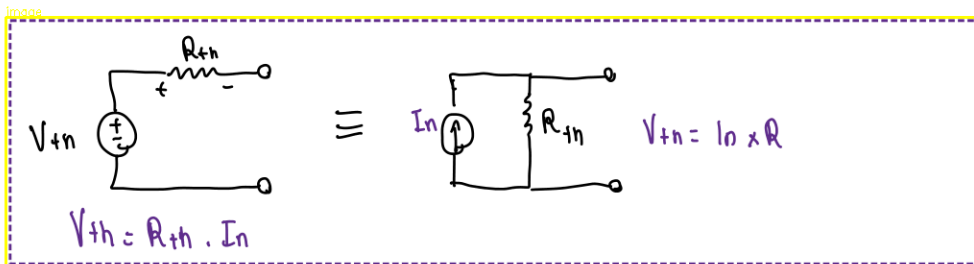
$$i = 15 \times \frac{8}{24+8} = 3,75 \text{ Amp}$$

$$\hookrightarrow V_A = 3,75 \times 12 = 45V$$

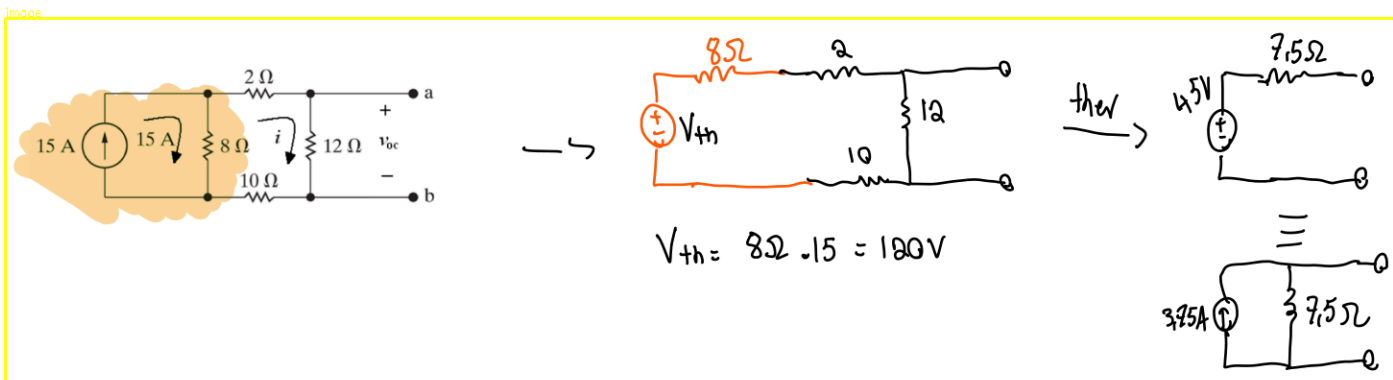


← circuit their equivalent

Norton



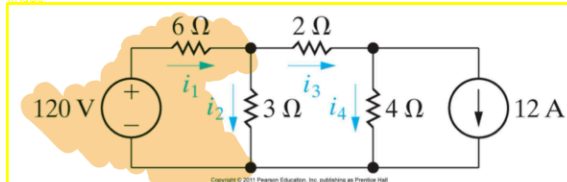
Same exo with Norton



Principe de superposition – Exemple (1)

- Calculer le courant i_3

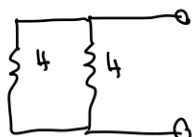
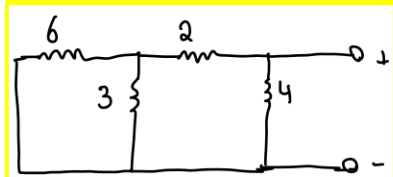
image



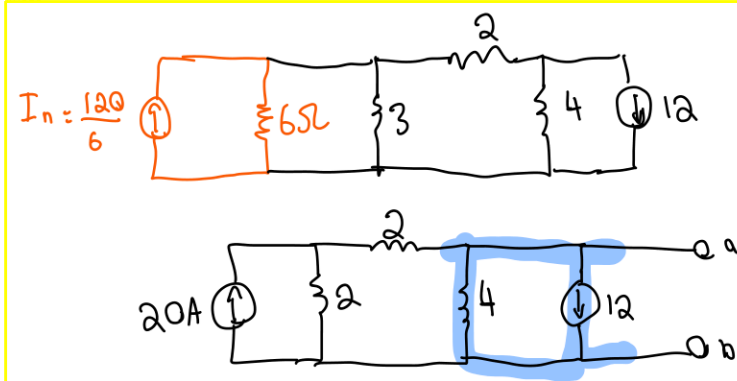
$$i_1 = i_3 + i_2$$

$$6i_1 =$$

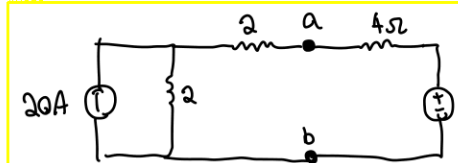
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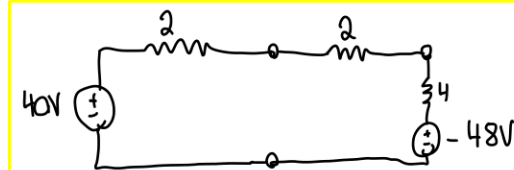
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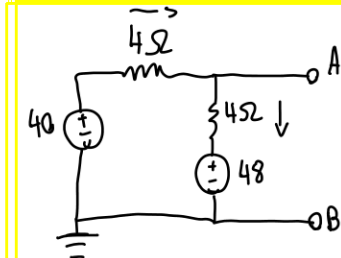
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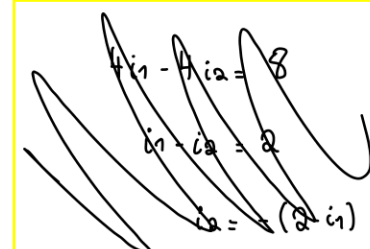
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image



image

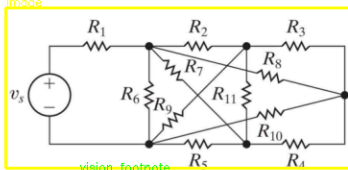
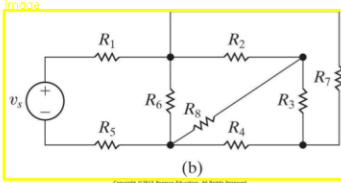
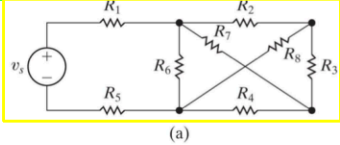


Vocabulaire:

- Noeud** : point où 2+ composants se rejoignent
- Noeud essentiel** : 3+ compo
- parcours** : chemin avec n° compo en double
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- Boucle** : parcours fermé
- Maille** : boucle sans autres boucle

Terminologie

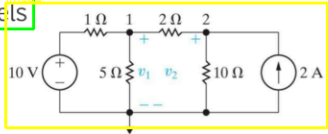
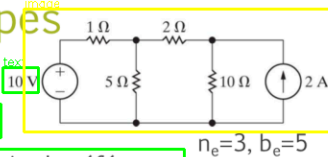
- Circuit planaire**: Circuit qui peut se dessiner sur un plan sans que des branches se croisent.



Méthode des nœuds - Étapes

Basée sur le LKC aux nœuds essentiels

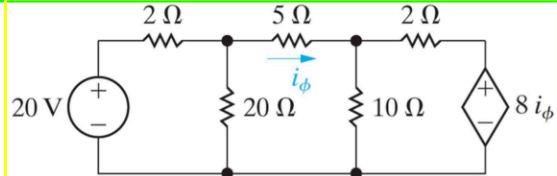
- Circuits planaires ou pas**: Redessiner clairement
- Sélectionner un des nœuds essentiels pour en faire la référence
- Nommer (variables) les tensions des nœuds essentiels restants
- Écrire les équations LKC des nœuds essentiels **restants** ($n_e - 1$ équations LKC)
 - Courants inconnus exprimés en fonction des tensions et résistances (éventuellement impédances)
- Solutionner pour les tensions de nœuds essentiels
- Vérifier votre solution
- Déduire d'autres quantités (courants) au besoin



Méthode spécifique à intra
maybe final too

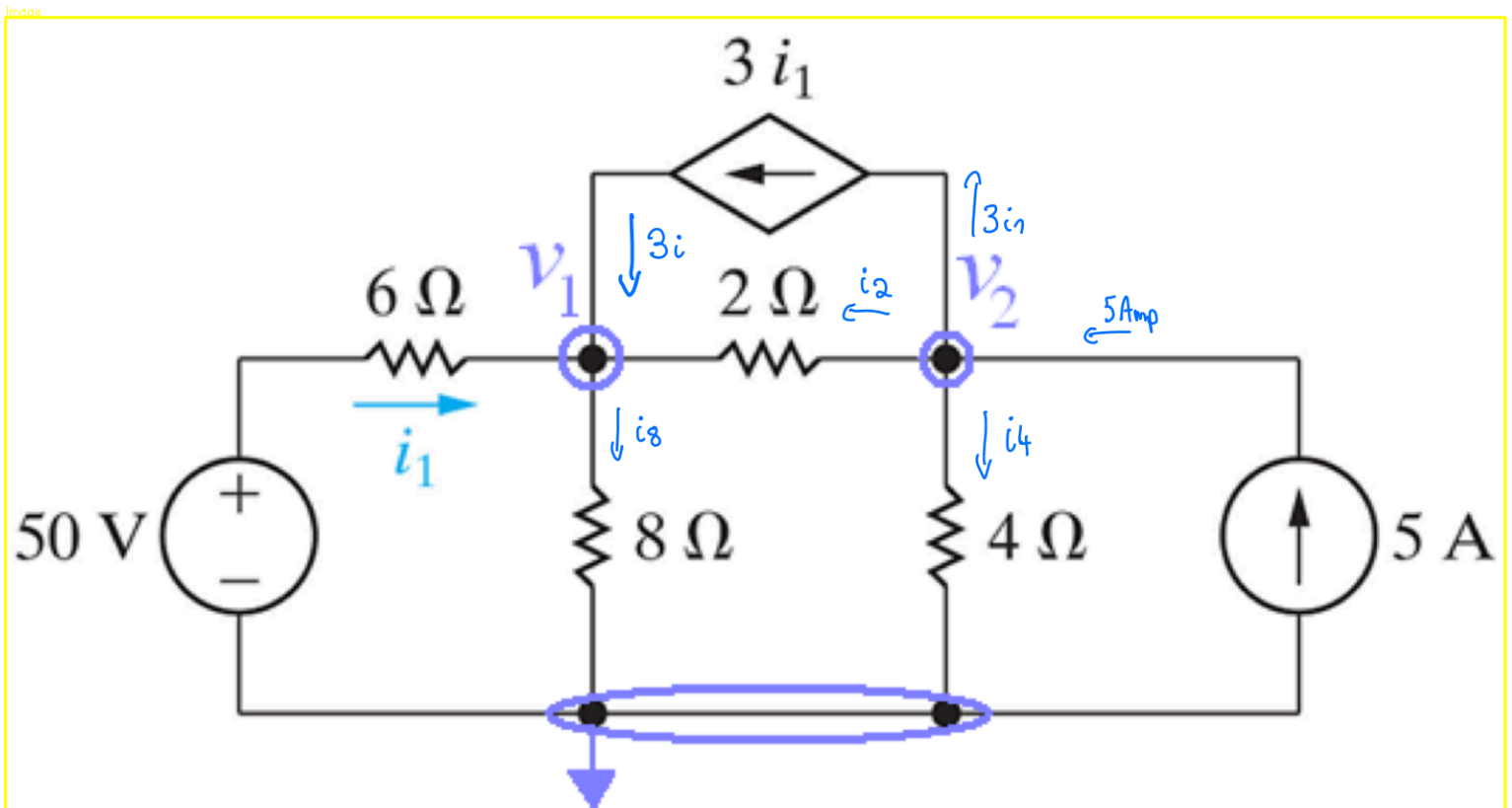
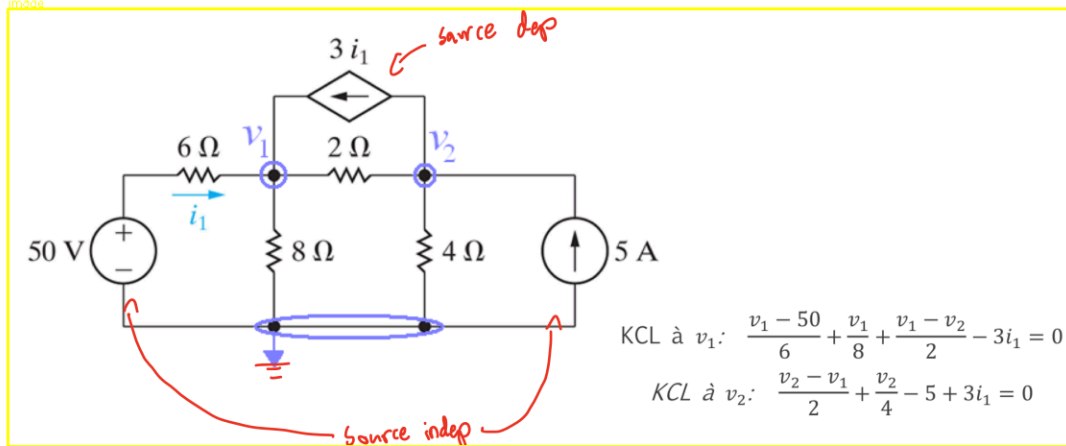
Cas spécial: Sources dépendantes

- La même méthode s'applique s'il y a des sources dépendantes
- Il faut une **équation de substitution** pour chaque source dépendante et qui définit la variable dont dépend la source dépendante **en termes d'autres variables existantes**
 - Cela permet de relier les deux quantités entre elles (donc une variable de source inconnue de moins)



Méthode des nœuds – Exemple 3 (1)

- Trouvez la puissance associée à chaque source.



$$\begin{aligned} V_1: & \quad 4i_1 + i_2 = i_3 \\ V_2: & \quad 5 = 3i_1 + i_2 + i_4 \end{aligned}$$

$$\begin{aligned} i_1 &= \frac{50 - V_1}{6} & i_2 &= \frac{V_2 - V_1}{2} \\ i_3 &= V_1/8 & i_4 &= V_2/4 \end{aligned}$$

$$\left. \begin{aligned} V_1: & \quad \frac{4}{6} (50 - V_1) + \frac{V_2 - V_1}{2} = \frac{V_1}{8} \\ V_2: & \quad 5 = \frac{3}{6} (50 - V_1) + \frac{V_2 - V_1}{2} + \frac{V_2}{4} \end{aligned} \right\} \rightarrow \text{Ti solve} \Rightarrow \begin{aligned} V_1 &= 30V \\ V_2 &= 16V \end{aligned}$$

display: formula

$$\left[\begin{aligned} i_1 &= \frac{50-32}{6} = 3A & i_2 &= \frac{32-16}{2} = 8A & i_4 &= 8A & i_8 &= 4A \end{aligned} \right.$$

$p = Vi$

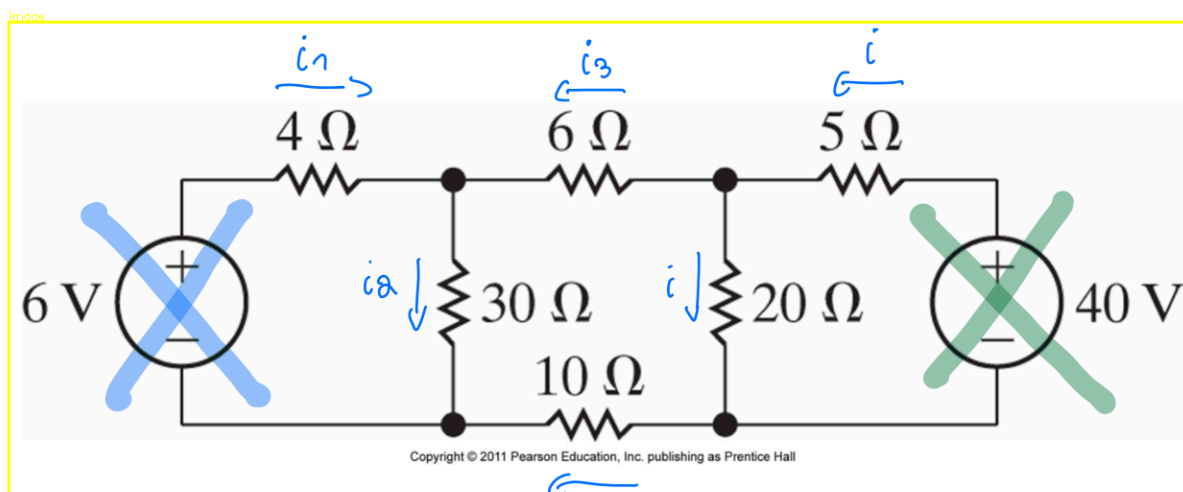
paragraph: title

Simplifier un circuit

text

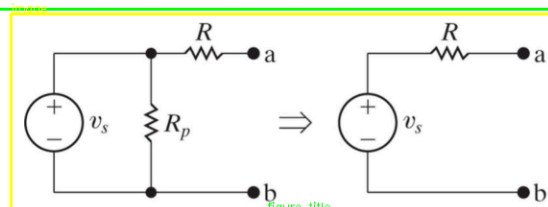
- Est-ce que la source de 6 volts absorbe ou délivre de la puissance?

$i' = i \text{ sans bleu} + i \text{ sans vert}$

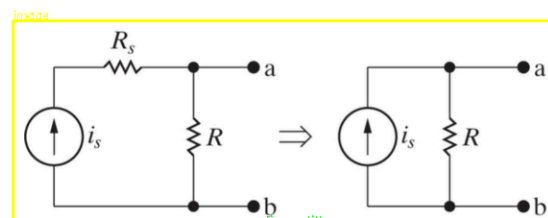


text

Cas spéciaux: on veut pouvoir faire les transformations suivantes



(a)



(b)